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Conference and Expo

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Making Bluetooth Connections

Tom Keel – Senior Software Engineer



Agenda

- What is Bluetooth?
- Demo!
- PalmOS Bluetooth Architecture & APIs
- Next Steps
- Questions and Answers

Bluetooth is not a dental disease. It's a ...

- low-cost,
- small form-factor,
- dynamic,
- secure,
- short-range, low-power,
- high-frequency, frequency-hopping,
- cable-replacement radio technology.



Not this!

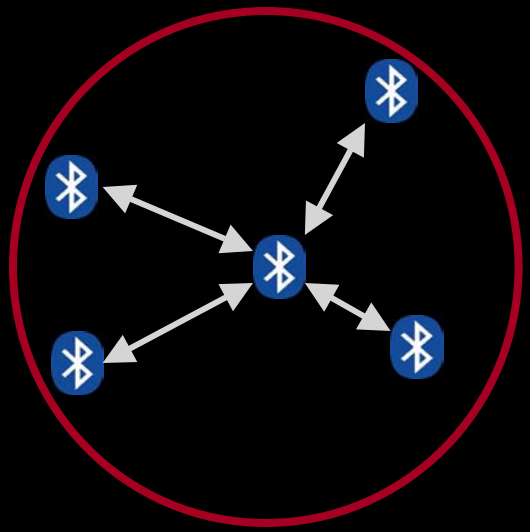
What a mouthful !

Bluetooth Standards

- Bluetooth SIG
  - Launched by Ericsson, IBM, Intel, Nokia, Toshiba
  - Currently 2000+ members
- Bluetooth Specification, revision 1.1
  - Core: protocols
  - Profiles: use cases
- Bluetooth Unplug Fests
  - Interoperability testing
  - Seven UPFs have been held to date
- Bluetooth Certification

Piconets

- A Piconet is 2 to 8 connected devices
- One device is master, all others are slaves
- All communication is through the master
- Connection initiator is master
- Master / slave switch possible



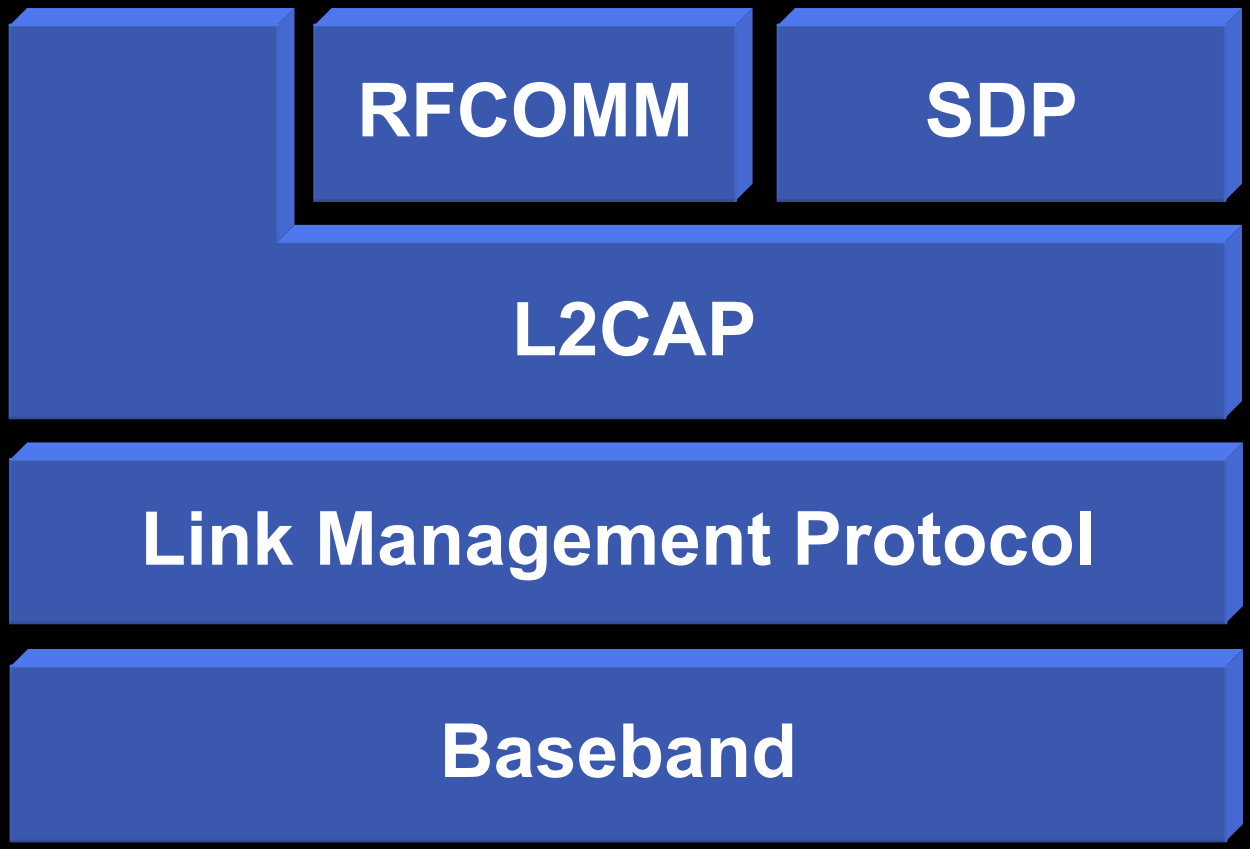
Bluetooth Addressing

- Each Bluetooth device is manufactured with a unique 48-bit Bluetooth device address.
- When a device becomes an active member of a piconet, it obtains a 3-bit address unique within the piconet.

Bluetooth Data Rate

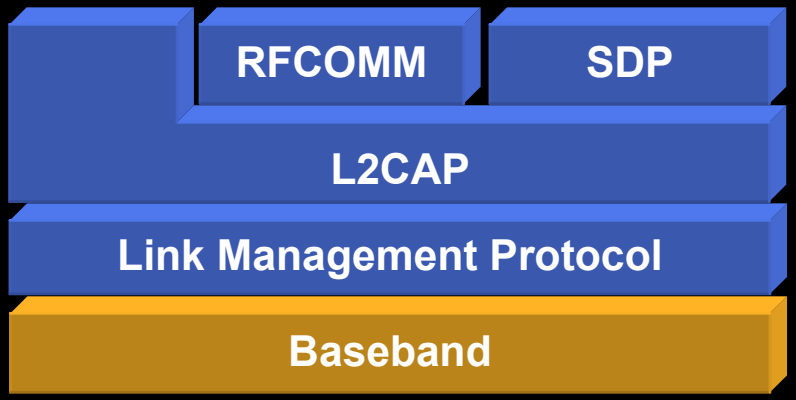
- Radio transmits at 1 Mb/s
- A timeslot is 625 microseconds
- Master talks for 1, 3, or 5 timeslots to 1 slave
- Slave answers in 1, 3, or 5 timeslots
- 1/3 of a timeslot lost when changing frequencies
- Maximum throughput of user data :
  - asymmetric: 723.2 kb/s & 57.6 kb/s return direction
  - symmetric: 433.9 kb/s both directions
- Up to 3 synchronous 64kb/s voice channels

Bluetooth Protocol Layering



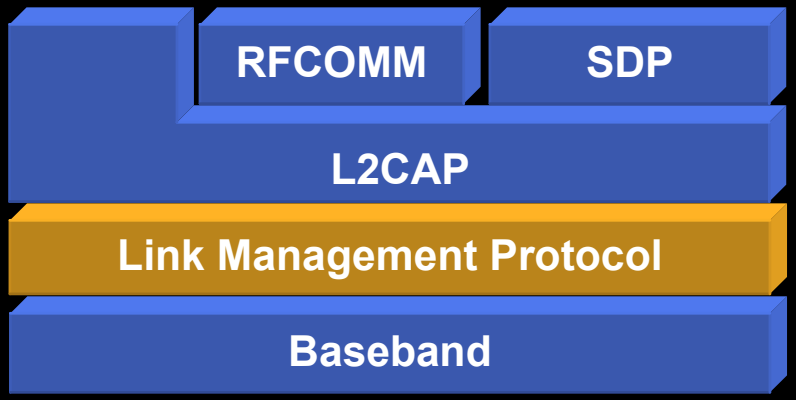
Baseband (aka Link Controller)

- Real-time radio control:
  - Timing
  - Hopping sequence
- Security calculations
- Maps ACL, SCO, etc. packets to the physical channel
- Reliability: Error detection, retransmission



Link Management Protocol

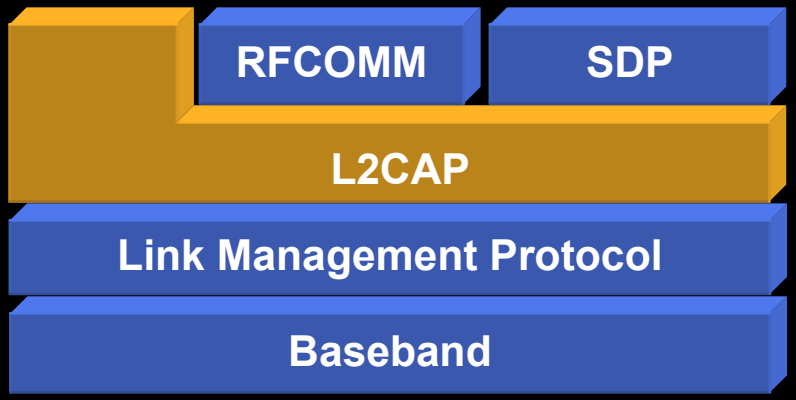
- Link managers talk to one another in order to establish, control, and destroy the ACL and SCO links. They say things like:



- "I want to be master!"
- "Prove you're who you claim to be!"
- "Let's use encryption!"

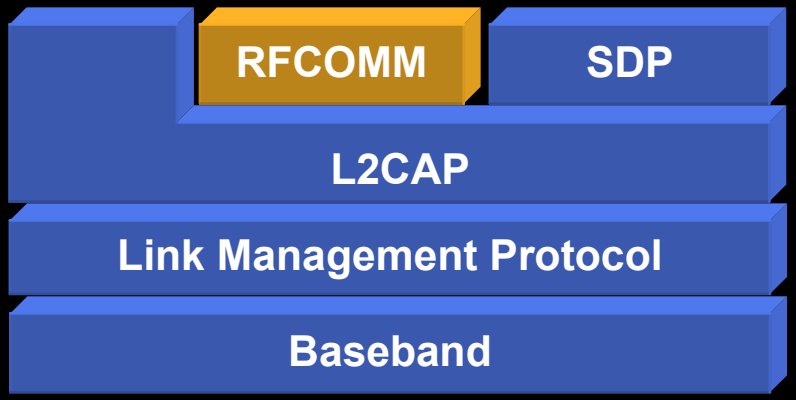
Logical Link Control and Adaptation Protocol

- Lowest layer useable by applications for data transfer
- Multiplexing : multiple L2CAP connections over one ACL link
- Segmentation and reassembly : big packets



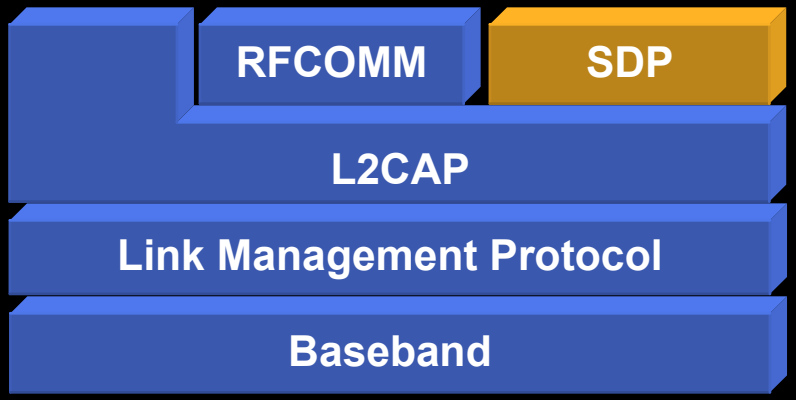
RfComm Protocol

- Multiplexing : multiple RfComm endpoints on one L2CAP endpoint
- Simulation of RS232 control lines
- Credit-based, end-to-end flow control



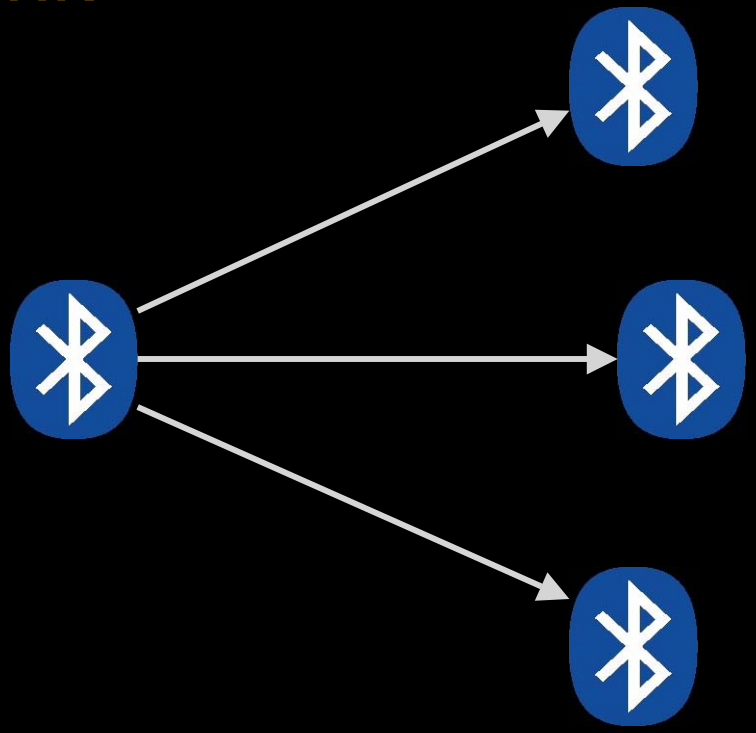
Service Discovery Protocol

- Servers manage a database describing the services they offer
- Clients use SDP to find the services advertised by servers



Establishing an ACL link

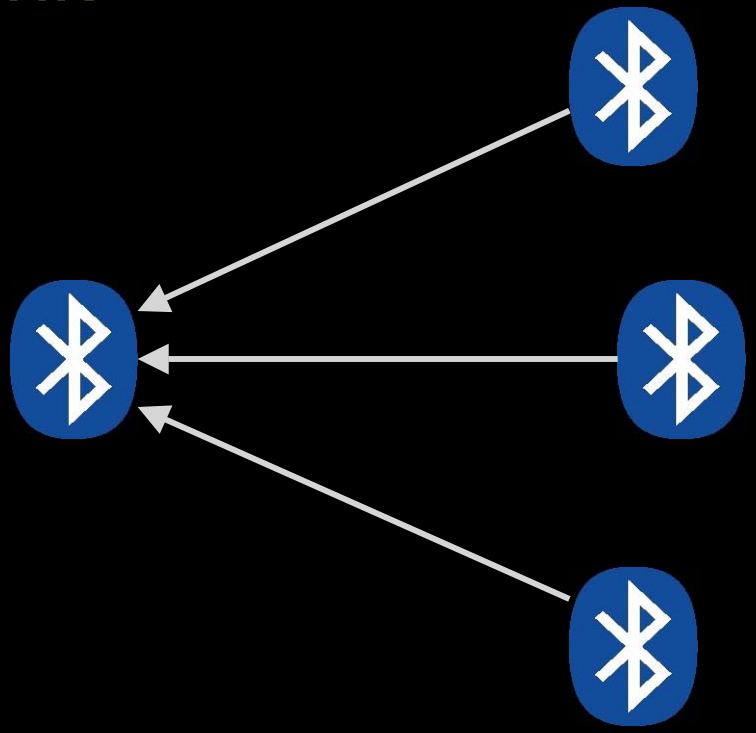
- **Inquiry**
- **Inquiry scanning**
- **Paging**
- **Page scanning**
- **Connected!**



Anyone out there?

Establishing an ACL link

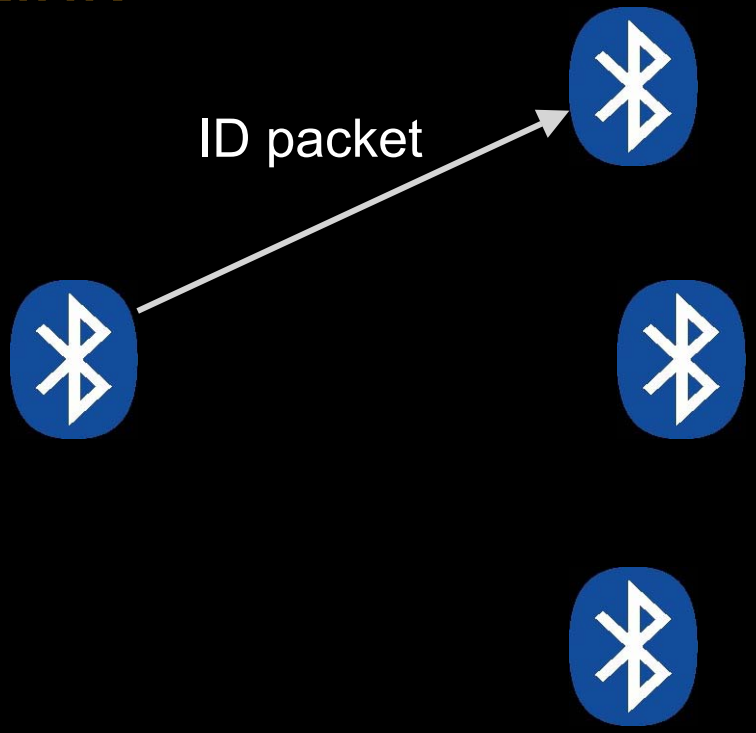
- Inquiry
- ⇨ Inquiry scanning
- Paging
- Page scanning
- Connected!



Yes, we are!

Establishing an ACL Link

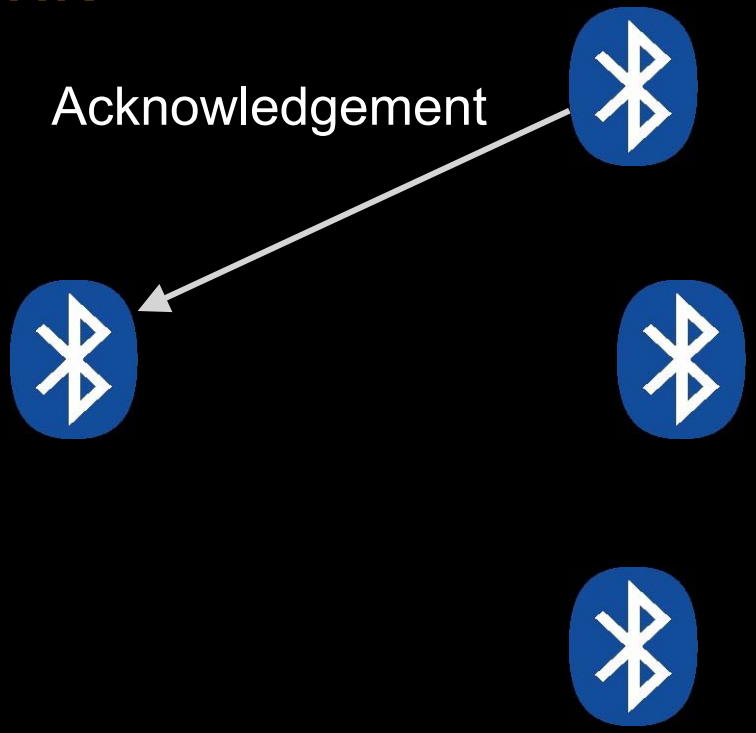
- **Inquiry**
- **Inquiry scanning**
- ⇨ **Paging**
- **Page scanning**
- **Connected!**



I'd like to connect to you …

Establishing an ACL link

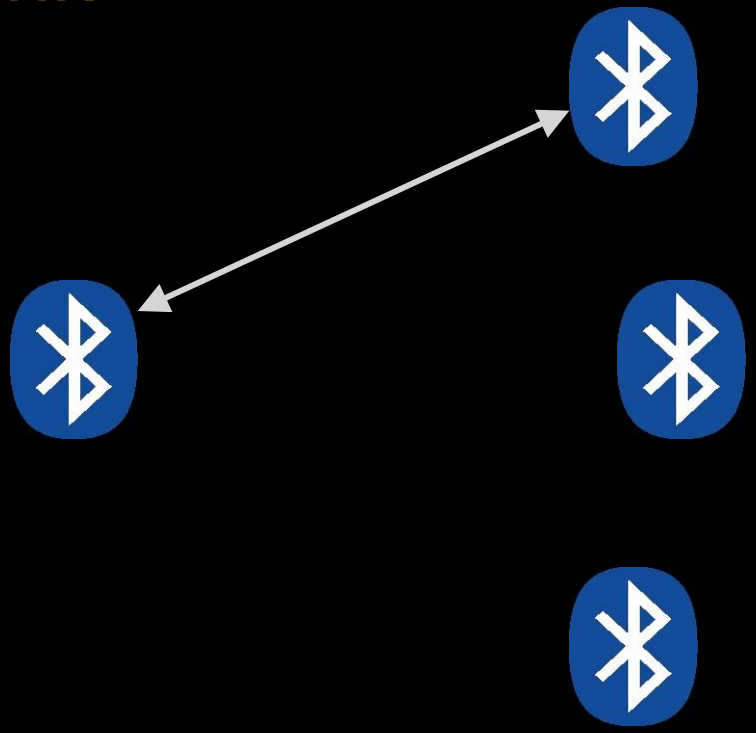
- **Inquiry**
- **Inquiry scanning**
- **Paging**
- ⇨ **Page scanning**
- **Connected!**



Sounds like a good idea!

Establishing an ACL link

- **Inquiry**
- **Inquiry scanning**
- **Paging**
- **Page scanning**
- ⇨ **Connected!**



Ready for SDP queries, L2CAP and RfComm connections …



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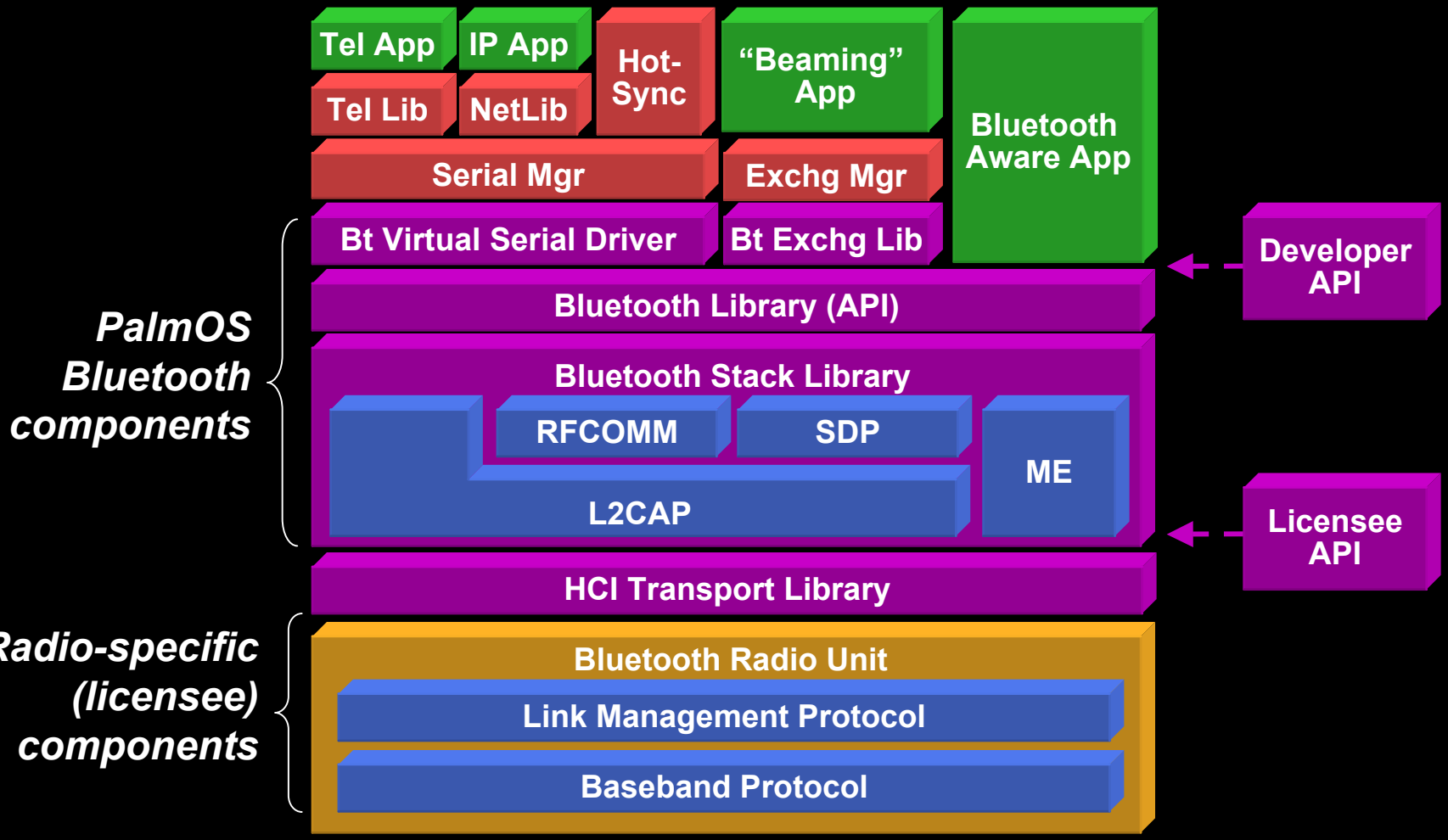
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Bluetooth Architecture



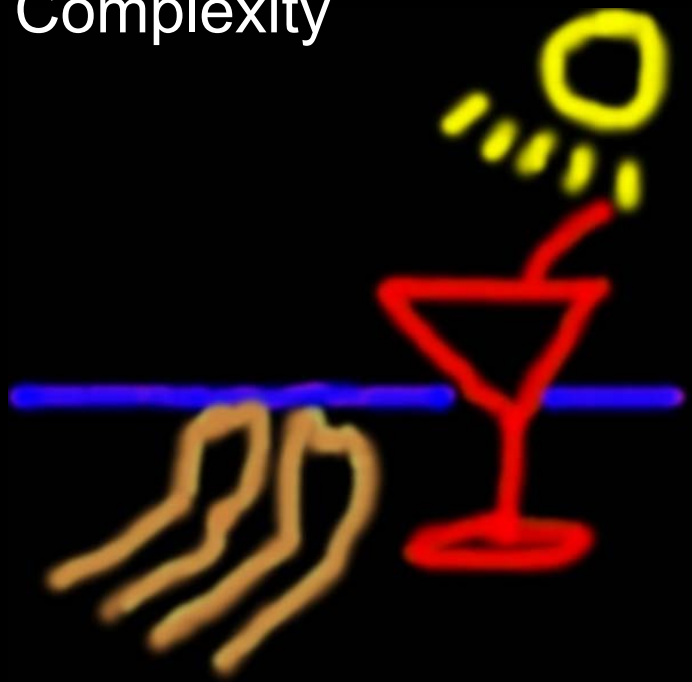
TCP/IP and Telephony Applications

- Power



Surf the net, dial your phone, ...

- Complexity



No changes needed

No Bluetooth knowledge required

Serial Applications

- Power



Cable replacement

- Complexity



Very little work

Applications using the Exchange Manager

- Power



Object exchange via OBEX

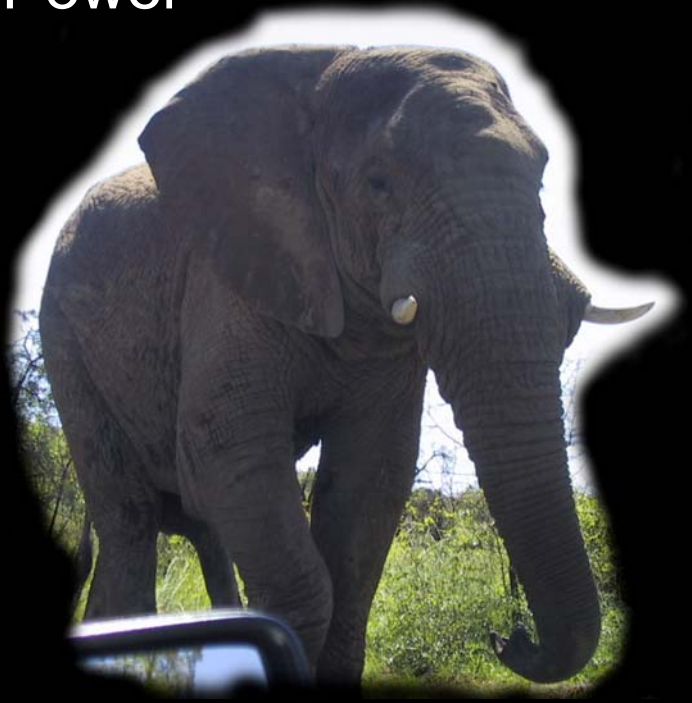
- Complexity



Minor changes to your app

Bluetooth Aware Applications

- Power



Very, very powerful!

- Complexity

$$\frac{x+y^2}{\sqrt{6-y^7}}$$

Easy for some people!

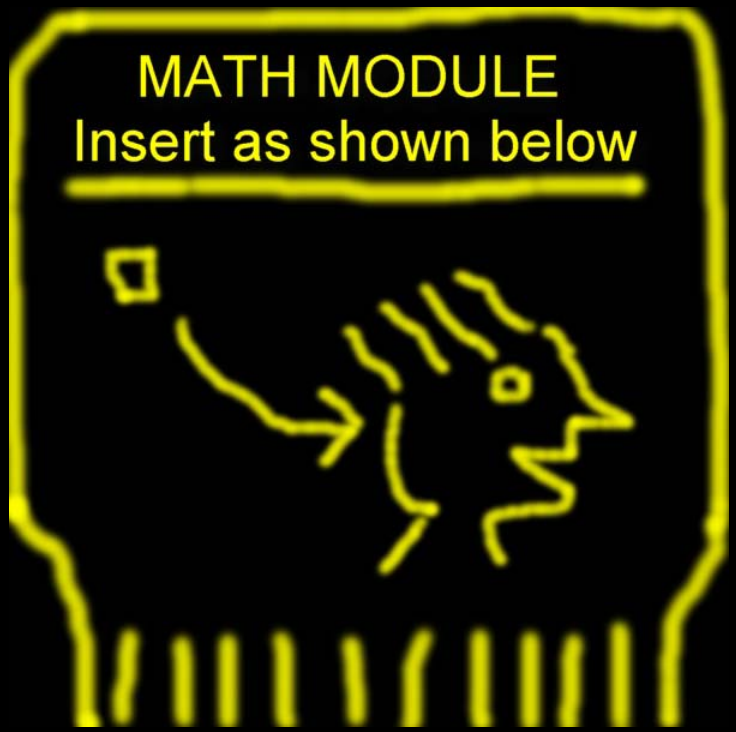
Fortunately …

- Sample Code makes things simple!

```
// Unlike ACL disconnects, closin
// will not result in a callback,
if( error == btLibErrNoError || e
    sdpSocket = INVALID_SOCKET;
else
{
    MyPrintF(gResultsForm, gResult
    PrintBtError(error);
}

// Create the L2CAP socket, and co
error = BtLibSocketCreate(gBtLibR
if (error == btLibErrNoError)
{
    connectInfo.remoteDeviceP = &g
    connectInfo.data.L2Cap.remote
    connectInfo.data.L2Cap.localM
    connectInfo.data.L2Cap.minRem
    error = BtLibSocketConnect( g
    if (error != btLibErrPending)
        BtLibSocketClose(gBtLibR
}
```

=



Serial Manager API

- Applications call SrmExtOpen()
  - Bluetooth-specific parameter block:
    - client / server
    - client : device address, service to look for
    - server : service to advertise
  - Client blocks until connected, runs progress dialog
  - Server returns immediately
- Legacy applications can call SrmOpen()
  - Client role assumed
  - Discovery performed
  - Looks for “Serial Port” service class

Serial Manager API - Client

- Can specify remote Bluetooth device address
  - If not given, a discovery will be performed
- Can specify a list of UUID's
  - Series of SDP queries will be made
  - First service class found will be used
  - UUID (128 bit) is "translated" into RFCOMM channel
- SrmExtOpen blocks until the RFCOMM channel is established
  - Progress dialog is displayed to the user

Serial Manager API - Server

- Specify Service Class UUID to advertise
  - If not, default Palm-specific UUID will be advertised
- Specify user-readable name for the service
- SrmExtOpen returns immediately
  - To wait for data call:
    - SrmReceive
    - SrmReceiveWait
    - SrmReceiveCheck

Exchange Manager API

- New Bluetooth Exchange library
  - Supports standard Exchange Manager API
  - Conforms to Object Push profile
  - Can send multiple objects to multiple destinations
- Can force Exchange Manager to use Bluetooth:
  - Send object(s) to single device
    - `_btobex://?_single/filename`
  - Send object(s) to multiple devices
    - `_btobex://?_multi/filename`
  - Send object(s) to specified device
    - `_btobex://address1[,address2, ...]/filename`

Exchange Manager API

- Can obtain the remote device address : call ExgControl() with exgLibCtlGetURL function code
- Tic-Tac-Toe sample code demonstrates this



Bluetooth API - Overview

- Many operations are asynchronous – the function you call returns immediately, and later you get an event in a callback
- No data buffering – left up to the application
- Basic functional groups in the API:
  - Management Entity
  - Sockets for L2CAP, RfComm, SDP
  - Service database management functions



Bluetooth API – M.E. Basic Functions

- Accessibility control (discoverable? connectable?)
- Inquiry and remoted device name retrieval
- Create / destroy ACL links
- Authentication / Encryption
- Master/Slave role switch

Bluetooth API – M.E. Utility Functions

- BtLibDiscoverMultipleDevices()
  - Caller specifies a set of Device Class UUIDS
  - Performs inquiry, name retrieval, service discovery
  - Progress and device selection UI done for you
  - Returns list of user-selected devices
- BtLibDiscoverSingleDevice()
  - Like DiscoverMultiple, but only one can be chosen
- Get user-friendly name of remote device
  - Caching scheme
  - temporary ACL connection if necessary

Bluetooth API - Management Entity Events

- BtLibManagementEventType
  - event code
  - supplementary info depending on event code
- Example Events
  - “radio state” + status
  - “response to inquiry” + BD_ADDR + Device Class
  - “ACL disconnection” + BD_ADDR
  - “accessibility change” + new accessibility mode

Bluetooth API - Sockets

- Sockets are opaque objects created with
  - type of socket (L2CAP, RfComm, SDP)
  - event handler and user context for it
- L2CAP and RfComm Sockets
  - communications endpoint
  - connection establishment / destruction
  - data transfer
- SDP Sockets
  - given a set of service classes, get handles on remote service records
  - given a handle and attribute type, get attribute value

Bluetooth API - Socket Events

- BtLibSocketEventType
  - event code
  - socket id
  - supplementary info depending on event code
- Examples
  - “inbound connection request” + BD_ADDR
  - “inbound connection accepted” + new socket id
  - “inbound data arrival” + data
  - “get service record handles complete” + list of service record handles

Bluetooth API - Hints

- This is PalmOS 4.x – event loop required.
- Don’t do data transfer in the callbacks. Instead:
  - Implement one or more queues for BT data events.
  - In the callbacks, enqueue the events, remembering to copy inbound data. If the BT queue was empty, post a nilEvent to the system event queue.
  - In the event loop, handle UI and system events first, then handle Bluetooth events.
- Close BtLib before exiting, but don’t unload it!



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Next Steps

- Download Bluetooth tools from the Resource Pavilion (www.palmos.com/bluetooth)
  - Bluetooth SDK
  - Sample code
  - White papers
- Purchase a Bluetooth hardware developer kit:

  Palm, Red-M, TDK, Northstar
- Join the Bluetooth forum
- Develop your Bluetooth application

Win a Bluetooth Developer Kit

Red-M Blade Palm OS® developer kit give-away

- 100 free kits for PalmSource delegates to win
- Visit the Red-M booth #116 to enter
  - Completed forms to be returned at Red- M booth
- Drawing will take place on Thursday 7th
  - Location – Developer Court, Civic Auditorium
  - Time - Lunch Break 1:00 pm



Making it easier to develop Bluetooth applications for Palm OS

red-M

Bluetooth™



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